

Neural Training

```
print('--- STEP 0: INSTALLING DEPENDENCIES ---')
# Install compatible versions of the required libraries for CUDA 13
# Remove conflicting versions
!pip uninstall -y transformers trl peft accelerate bitsandbytes datasets

# Install compatible versions
!pip install -q \
transformers==4.41.2 \
trl==0.9.6 \
peft==0.11.1 \
accelerate==0.31.0 \
datasets==2.20.0 \
-q bitsandbytes==0.48.1 triton

--- STEP 0: INSTALLING DEPENDENCIES ---
Found existing installation: transformers 4.41.2
Uninstalling transformers-4.41.2:
  Successfully uninstalled transformers-4.41.2
Found existing installation: trl 0.9.6
Uninstalling trl-0.9.6:
  Successfully uninstalled trl-0.9.6
Found existing installation: peft 0.11.1
Uninstalling peft-0.11.1:
  Successfully uninstalled peft-0.11.1
Found existing installation: accelerate 0.31.0
Uninstalling accelerate-0.31.0:
  Successfully uninstalled accelerate-0.31.0
Found existing installation: bitsandbytes 0.43.1
Uninstalling bitsandbytes-0.43.1:
  Successfully uninstalled bitsandbytes-0.43.1
Found existing installation: datasets 2.20.0
Uninstalling datasets-2.20.0:
  Successfully uninstalled datasets-2.20.0
  60.1/60.1 MB 14.7 MB/s eta 0:00:00
```

```
import transformers, trl, peft, accelerate, bitsandbytes, datasets, torch
```

```
print("transformers:", transformers.__version__)
print("trl:", trl.__version__)
print("peft:", peft.__version__)
print("accelerate:", accelerate.__version__)
print("bitsandbytes:", bitsandbytes.__version__)
print("datasets:", datasets.__version__)
print("torch:", torch.__version__)
print("cuda:", torch.version.cuda)
```

```
transformers: 4.41.2
trl: 0.9.6
peft: 0.11.1
accelerate: 0.31.0
bitsandbytes: 0.48.1
datasets: 2.20.0
torch: 2.11.0+cu128
cuda: 12.8
```

```
import torch
import pandas as pd
from datasets import Dataset
from transformers import AutoModelForCausalLM, AutoTokenizer, BitsAndBytesConfig, TrainingArguments
from peft import LoraConfig, get_peft_model, prepare_model_for_kbit_training
from trl import SFTTrainer
```

```
print("--- STEP 1: MERGING DATASETS ---")
# Load the cleaned datasets you created earlier
codex_df = pd.read_csv("codex_clean.csv")
mqa_df = pd.read_csv("unified_mqa_train.csv")
```

```
--- STEP 1: MERGING DATASETS ---
```

```
master_train_df = pd.concat([codex_df, mqa_df], ignore_index=True)
master_train_df = master_train_df.sample(frac=1, random_state=42).reset_index(drop=True) # Shuffle
dataset = Dataset.from_pandas(master_train_df)
```

```
model_id = "microsoft/Phi-3-mini-4k-instruct"
tokenizer = AutoTokenizer.from_pretrained(model_id, trust_remote_code=True)
tokenizer.pad_token = tokenizer.eos_token
```

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF\_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (<https://huggingface.co/settings/tokens>), set
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.

```
warnings.warn(
tokenizer_config.json:      3.44k/? [00:00<00:00, 142kB/s]

tokenizer.model: 100%                               500k/500k [00:01<00:00, 437kB/s]

tokenizer.json:      1.94M/? [00:00<00:00, 21.2MB/s]

added_tokens.json: 100%                             306/306 [00:00<00:00, 8.11kB/s]

special_tokens_map.json: 100%                       599/599 [00:00<00:00, 21.3kB/s]

Special tokens have been added in the vocabulary, make sure the associated word embeddings are fine-tuned or trained.
```

```
def format_prompt(row):
    """The strict format we force the AI to learn."""
    prompt = f"### Instruction: Write Python code to solve the math problem. Store the answer in 'result'.\\n"
    prompt += f"### Question:\\n{row['question']}\\n"
    prompt += f"### Code:\\n{row['code']}\\n"
    return {"text": prompt + tokenizer.eos_token}
```

```
formatted_dataset = dataset.map(format_prompt)
```

```
Map: 100%                                           12000/12000 [00:02<00:00, 5336.44 examples/s]
```

```
print("--- STEP 2: LOADING MODEL IN 4-BIT ---")
bnb_config = BitsAndBytesConfig(
    load_in_4bit=True,
    bnb_4bit_use_double_quant=True,
    bnb_4bit_quant_type="nf4",
    bnb_4bit_compute_dtype=torch.bfloat16
)
```

```
--- STEP 2: LOADING MODEL IN 4-BIT ---
```

```
# !pip uninstall -y numpy pandas
# !pip install -q numpy==1.26.4 pandas==2.2.2
```

```
model = AutoModelForCausalLM.from_pretrained(
    model_id, quantization_config=bnb_config, device_map="auto", trust_remote_code=True
)
model = prepare_model_for_kbit_training(model)
```

```
config.json: 100%                               967/967 [00:00<00:00, 41.7kB/s]

configuration_phi3.py:      11.2k/? [00:00<00:00, 311kB/s]
A new version of the following files was downloaded from https://huggingface.co/microsoft/Phi-3-mini-4k-instruct:
- configuration_phi3.py
. Make sure to double-check they do not contain any added malicious code. To avoid downloading new versions of the code file
modeling_phi3.py:      73.2k/? [00:00<00:00, 3.48MB/s]
A new version of the following files was downloaded from https://huggingface.co/microsoft/Phi-3-mini-4k-instruct:
- modeling_phi3.py
. Make sure to double-check they do not contain any added malicious code. To avoid downloading new versions of the code file
WARNING:transformers_modules.microsoft.Phi-3-mini-4k-instruct.f39ac1d28e925b323eae81227eaba4464caced4e.modeling_phi3:flash-
WARNING:transformers_modules.microsoft.Phi-3-mini-4k-instruct.f39ac1d28e925b323eae81227eaba4464caced4e.modeling_phi3:Current
model.safetensors.index.json:      16.5k/? [00:00<00:00, 689kB/s]

Downloading shards: 100%                           2/2 [01:00<00:00, 29.22s/it]

model-00001-of-00002.safetensors: 100%             4.97G/4.97G [00:33<00:00, 168MB/s]

model-00002-of-00002.safetensors: 100%             2.67G/2.67G [00:25<00:00, 102MB/s]

Loading checkpoint shards: 100%                     2/2 [00:36<00:00, 17.19s/it]
/usr/local/lib/python3.12/dist-packages/bitsandbytes/backends/cuda/ops.py:212: FutureWarning: _check_is_size will be removed
torch._check_is_size(blocksize)
generation_config.json: 100%                       181/181 [00:00<00:00, 14.5kB/s]
```

```
print("--- STEP 3: APPLYING QLoRA ---")
peft_config = LoraConfig(
```

```
    r=16,
    lora_alpha=32,
    target_modules="all-linear",
    lora_dropout=0.05,
    bias="none",
    task_type="CAUSAL_LM"
)
model = get_peft_model(model, peft_config)
```

--- STEP 3: APPLYING QLoRA ---

```
print("--- STEP 4: TRAINING ---")
training_args = TrainingArguments(
    output_dir="./phi3-math-agent",
    per_device_train_batch_size=2,
    gradient_accumulation_steps=4,
    learning_rate=2e-4,
    max_steps=500, # Use max_steps for a quick test, or change to num_train_epochs=1 for full training
    fp16=True,
    optim="paged_adamw_8bit"
)
```

--- STEP 4: TRAINING ---

```
trainer = SFTTrainer(
    model=model,
    train_dataset=formatted_dataset,
    peft_config=peft_config,
    dataset_text_field="text",
    max_seq_length=512,
    tokenizer=tokenizer,
    args=training_args
)

trainer.train()
```

```

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_deprecation.py:100: FutureWarning: Deprecated argument(s) use
Deprecated positional argument(s) used in SFTTrainer, please use the SFTConfig to set these arguments instead.
  warnings.warn(message, FutureWarning)
/usr/local/lib/python3.12/dist-packages/trl/trainer/sft_trainer.py:280: UserWarning: You passed a `max_seq_length` argument
  warnings.warn(
/usr/local/lib/python3.12/dist-packages/trl/trainer/sft_trainer.py:318: UserWarning: You passed a `dataset_text_field` argum
  warnings.warn(

Map: 100%                               12000/12000 [00:06<00:00, 1536.69 examples/s]

/usr/local/lib/python3.12/dist-packages/trl/trainer/sft_trainer.py:408: UserWarning: You passed a tokenizer with `padding_si
  warnings.warn(
/usr/local/lib/python3.12/dist-packages/accelerate/accelerator.py:477: FutureWarning: `torch.cuda.amp.GradScaler(args...)` i
  self.scaler = torch.cuda.amp.GradScaler(**kwargs)
max_steps is given, it will override any value given in num_train_epochs
wandb: WARNING The `run_name` is currently set to the same value as `TrainingArguments.output_dir`. If this was not intended
/usr/local/lib/python3.12/dist-packages/notebook/notebookapp.py:191: SyntaxWarning: invalid escape sequence `\'\'
  | _ | | ' _ \ \ _ ` / _ ` | _/ -_)
wandb: (1) Create a W&B account
wandb: (2) Use an existing W&B account
wandb: (3) Don't visualize my results
wandb: Enter your choice: 1
wandb: You chose 'Create a W&B account'
wandb: Create an account here: https://wandb.ai/authorize?signup=true&ref=models
wandb: After creating your account, create a new API key and store it securely.
wandb: Paste your API key and hit enter: .....
wandb: No netrc file found, creating one.
wandb: Appending key for api.wandb.ai to your netrc file: /root/.netrc
wandb: Currently logged in as: ekta418214 (ekta418214-thadomal-shahani-engineering-college) to https://api.wandb.ai. Use `wa
Tracking run with wandb version 0.27.2
Run data is saved locally in /content/wandb/run-20260614_103454-u0zh5b4c
Syncing run \_phi3-math-agent to Weights & Biases \(docs\)
View project at https://wandb.ai/ekta418214-thadomal-shahani-engineering-college/huggingface
View run at https://wandb.ai/ekta418214-thadomal-shahani-engineering-college/huggingface/runs/u0zh5b4c
WARNING:transformers_modules.microsoft.Phi-3-mini-4k-instruct.f39ac1d28e925b323eae81227eaba4464caced4e.modeling_phi3:`use_ca
/usr/local/lib/python3.12/dist-packages/torch/_dynamo/eval_frame.py:1263: UserWarning: torch.utils.checkpoint: the use_reent
  return fn(*args, **kwargs)

```